File Signature Detection using Python

Python Program:

```
# File signatures (magic numbers)
SIGNATURES = {
  b"\xFF\xD8\xFF": "JPEG image",
  b"\x89PNG": "PNG image",
  b"%PDF": "PDF document",
  b"PK\x03\x04": "ZIP archive (or .docx/.xlsx)",
}

def identify_file(path):
  with open(path,"rb") as f:
    header=f.read(8)
  for sig,filetype in SIGNATURES.items():
    if header.startswith(sig):
      return filetype
  return "Unknown file type"

with open("photo.txt","wb") as f:
  f.write(b"\xFF\xD8\xFF\xE0"+b"\x00"*20)
with open("document.dat","wb") as f:
  f.write(b"%PDF-1.4"+b"\x00"*20)

for filename in ["photo.txt","document.dat"]:
  print(f"{filename:15s} -> Actual type: {identify_file(filename)}")
```

Sample Output:

photo.txt   -> Actual type: JPEG image
document.dat -> Actual type: PDF document

Explanation:

This program identifies the real type of a file by reading its file signature (magic number) instead of relying on the file extension. It compares the first few bytes of each file with known signatures for JPEG, PNG, PDF, and ZIP files. Even if a file is renamed with a different extension, the program correctly detects its actual file type.